

Amazon Egypt Data Warehouse Project

An end-to-end Data Warehouse built from scratch on SQL Server, simulating a real e-commerce marketplace — from raw ingestion to a business-ready Galaxy Schema.

19

Tables (Gold Layer)

6 / 13

Fact / Dimension Tables

3

Layers — Bronze / Silver / Gold

Jan 2025

– Sep 2026 Data Range

What this project simulates

- **Egyptian e-commerce marketplace** — both first-party (Amazon Retail) and third-party sellers, with commission on marketplace orders.
- **Full order lifecycle** — sales, returns, reviews, delivery performance, and inventory stock levels.
- **Marketing attribution** — which platform / campaign / adset a customer's order came from.
- **Seasonal demand** — real Egyptian campaigns (Ramadan, Eid al-Fitr, Eid al-Adha, Mother's Day, White Friday) with a measurable lift on sales.
- **Customer segmentation left un-baked** — RFM is computed live with SQL, not pre-calculated in the dimension.

~41K

Sales order lines

~28K

Orders

80

Sellers (1 retail + 79 marketplace)

300

Products

Bronze → Silver → Gold

01 · BRONZE

Raw Ingestion

All 19 CSV source files loaded as-is via **BULK INSERT**, every column **NVARCHAR**. No constraints, no cleaning — a faithful copy of the source data.



02 · SILVER

Cleaned & Typed

Correct data types via **TRY_CAST**, NULL handling with **NULLIF**, text-to-BIT conversion. Primary Keys only — no Foreign Keys yet, to keep loads fast and flexible.



03 · GOLD

Business-Ready

Calculated columns (**profit**, **CTR**, **days_of_supply**, **delivery duration**), full **Primary + Foreign Keys** for referential integrity — ready for BI / analytics.

Loading raw data with BULK INSERT

Every source CSV is loaded into an all-NVARCHAR staging table. A UTF-8 / LF-terminated file format required explicit ROWTERMINATOR='0x0a' and CODEPAGE='65001'.

```

1
2
3  -- 1) Schemas (one per layer)
4
5  IF NOT EXISTS (SELECT 1 FROM sys.schemas WHERE name = 'bronze') EXEC('CREATE SCHEMA bron:
6  IF NOT EXISTS (SELECT 1 FROM sys.schemas WHERE name = 'silver') EXEC('CREATE SCHEMA silv
7  IF NOT EXISTS (SELECT 1 FROM sys.schemas WHERE name = 'gold') EXEC('CREATE SCHEMA gold
8  GO
9
10
11  -- 2) Bronze - Dimension tables
12
13  DROP TABLE IF EXISTS bronze.dim_date;
14  CREATE TABLE bronze.dim_date (
15      date_id          NVARCHAR(50),
16      full_date         NVARCHAR(50),
17      [day]             NVARCHAR(50),
18      [month]           NVARCHAR(50),
19      month_name        NVARCHAR(50),
20      [quarter]         NVARCHAR(50),
21      [year]            NVARCHAR(50),
22      day_name          NVARCHAR(50),
23      is_weekend        NVARCHAR(50),
24      is_holiday_season NVARCHAR(50),
25      holiday_name      NVARCHAR(100)
26  );
27
28  DROP TABLE IF EXISTS bronze.dim_customer;
29  CREATE TABLE bronze.dim_customer (
30      customer_id       NVARCHAR(50),
31      full_name          NVARCHAR(200),

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8
Disconnected.

Bronze schema & staging tables

```

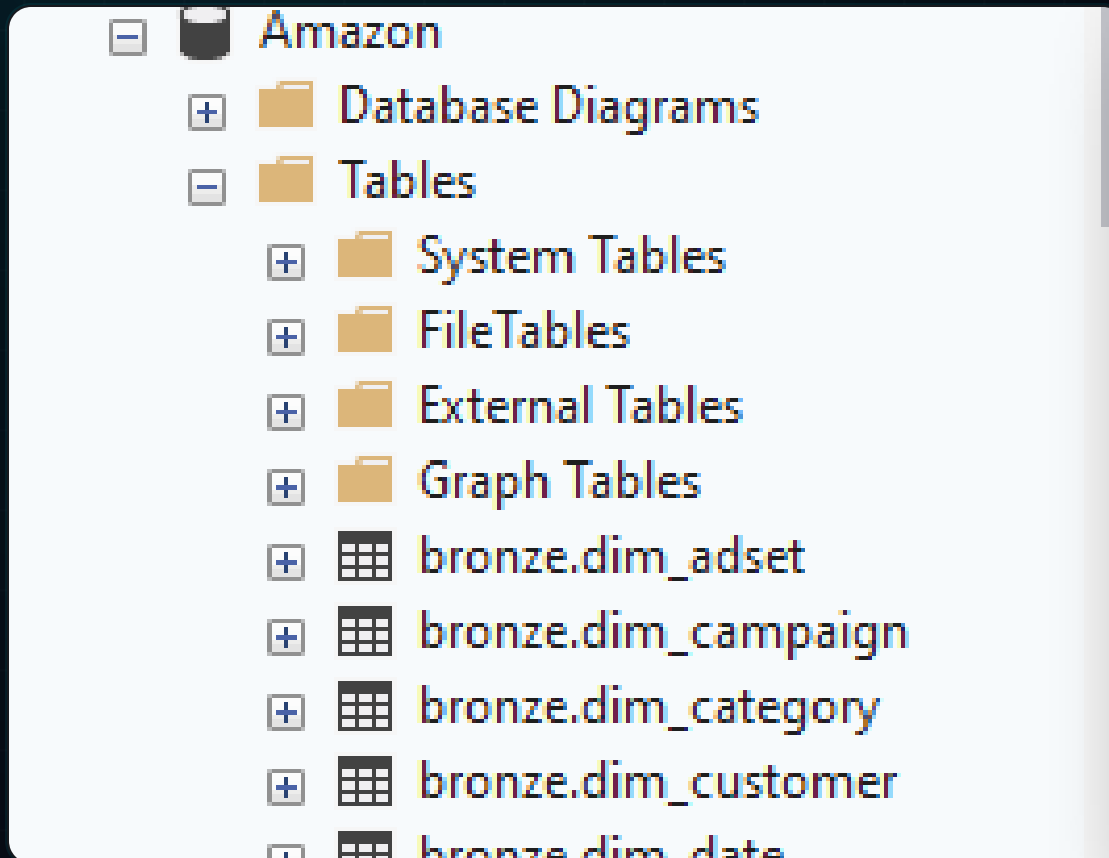
1  CREATE OR ALTER PROCEDURE bronze.BulkInsert AS
2  BEGIN
3
4      -- dim_date
5      TRUNCATE TABLE bronze.dim_date;
6
7      BULK INSERT bronze.dim_date
8      FROM 'D:\Big Projects\Amazon\amazon_egypt_dwh_dataset\dim_date.csv'
9      WITH (
10         FORMAT          = 'CSV',
11         FIRSTROW        = 2,
12         FIELDTERMINATOR = ',',
13         ROWTERMINATOR   = '0x0a',
14         CODEPAGE         = '65001',
15         TABLOCK
16     );
17
18
19     -- dim_customer
20     TRUNCATE TABLE bronze.dim_customer;
21
22     BULK INSERT bronze.dim_customer
23     FROM 'D:\Big Projects\Amazon\amazon_egypt_dwh_dataset\dim_customer.csv'
24     WITH (
25         FORMAT          = 'CSV',
26         FIRSTROW        = 2,
27         FIELDTERMINATOR = ',',
28         ROWTERMINATOR   = '0x0a',
29         CODEPAGE         = '65001',
30         TABLOCK
31     );

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

BULK INSERT routine

Verifying the raw load



All bronze tables populated

```
);  
  
DROP TABLE IF EXISTS bronze.dim_customer;  
CREATE TABLE bronze.dim_customer (  
    customer_id      NVARCHAR(50),  
    full_name        NVARCHAR(200),  
    gender           NVARCHAR(50),  
    birth_date       NVARCHAR(50),  
    governorate      NVARCHAR(100),  
    city             NVARCHAR(100),  
    registration_date NVARCHAR(50),  
    membership_tier  NVARCHAR(50),  
    acquisition_channel NVARCHAR(100)  
);  
  
DROP TABLE IF EXISTS bronze.dim_product;  
CREATE TABLE bronze.dim_product (  
    product_id      NVARCHAR(50),  
    category_id     NVARCHAR(50),  
    product_name    NVARCHAR(300),  
    brand           NVARCHAR(100),  
    unit_cost       NVARCHAR(50),  
    list_price      NVARCHAR(50),  
    launch_date     NVARCHAR(50),  
    primary_seller_id NVARCHAR(50)  
);
```

bronze.dim_customer – raw rows

Cleaning & Transformation

- **TRY_CAST** every column into its correct type — fails to NULL instead of breaking the whole batch.
- **NULLIF('', ...)** turns blank strings into true NULLs before casting.
- **CASE WHEN** pattern converts text 'True'/'False' values into real **BIT** columns.
- Primary Keys defined here — Foreign Keys deliberately deferred to Gold.
- A NULL-count check closes every table script before moving to the next one.

```
1  /* Create silver dim Adset table */
2  Drop table if exists Silver.dim_adset;
3  CREATE TABLE Silver.dim_adset (
4      adset_id int PRIMARY KEY Not Null,
5      campaign_id int Not Null,
6      adset_name nvarchar(255) Null,
7      target_age_group nvarchar(255) Null,
8      target_gender nvarchar(255) Null ,
9
10 );
11 go
12
13 /* Insert data after cleaning and transformation data type*/
14 Insert into Silver.dim_adset (adset_id, campaign_id, adset_name, target_age_group, target_gender)
15 SELECT
16     Try_CAST(adset_id AS int) AS adset_id,
17     Try_CAST(campaign_id AS int) AS campaign_id,
18     adset_name,
19     target_age_group,
20     target_gender
21 FROM bronze.dim_adset;
22
23 /*Check for any null values in the primary key column*/
24 Select * from Silver.dim_adset;
25 SELECT COUNT(*) FROM silver.dim_adset;
26 SELECT * FROM silver.dim_adset WHERE adset_id IS NULL OR campaign_id IS NULL;
```

silver.dim_adset — typed & cleaned

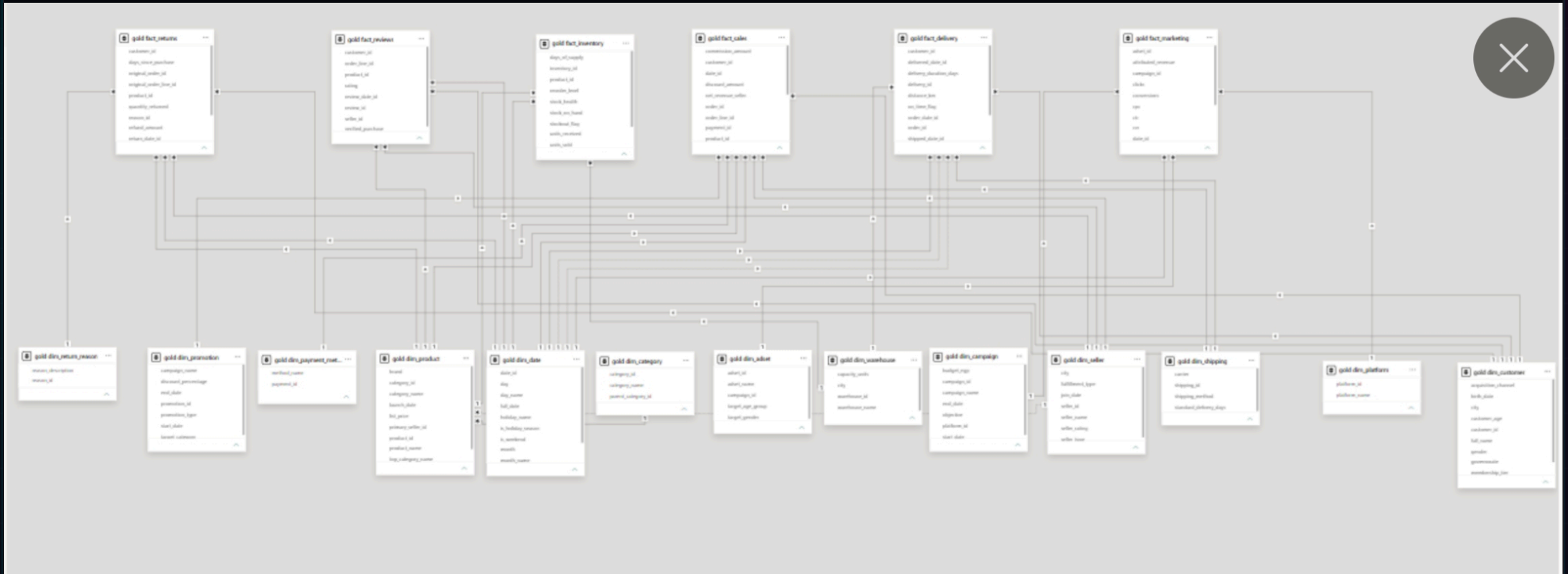
Calculated columns & constraints

- **fact_sales** — total_cost, profit, profit_margin_pct, net_revenue_seller.
- **fact_marketing** — CTR, CVR, CPC, ROAS (with divide-by-zero guards).
- **fact_inventory** — days_of_supply, stock_health.
- **fact_delivery** — delivery_duration_days, on_time_flag.
- **Full PK + FK constraints** defined here — guarantees referential integrity before the data reaches BI.

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Galaxy Schema — 6 Fact Tables, 13 Dimensions



Full ERD — fact_sales, fact_returns, fact_reviews, fact_marketing, fact_inventory, fact_delivery, each standing independently and sharing conformed dimensions.

THANK YOU

Built end-to-end on **SQL Server**

Bronze · Silver · Gold — from raw CSVs to a business-ready Galaxy Schema, designed and implemented hands-on as a portfolio Data Warehouse project.

19

Tables

6

Fact
Tables

13

Dimensions